

Math 241, F18
Quiz 12

Name:

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Using the cylindrical coordinates to compute the triple integral $\iiint_E z\sqrt{x^2+y^2} dV$ where E is the solid that lies between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$, and between the planes $z = -2$ and $z = 4$.



$$E = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, \\ 1 \leq r \leq 3, \\ -2 \leq z \leq 4\}$$

$$\begin{aligned} \iiint_E z\sqrt{x^2+y^2} dV &= \int_0^{2\pi} \int_1^3 \int_{-2}^4 r z \cdot r dz dr d\theta \\ &= \int_0^{2\pi} \int_1^3 r^2 \left[\frac{z^2}{2} \right]_{z=-2}^{z=4} dr d\theta \\ &= \int_0^{2\pi} \int_1^3 6r^2 dr d\theta \\ &= \int_0^{2\pi} \left[2r^3 \right]_{r=1}^{r=3} d\theta \\ &= \int_0^{2\pi} 52 d\theta = 104\pi \end{aligned}$$

Problem 2. (5 points) Evaluate $\iiint_E \sqrt{x^2+y^2+z^2} dV$ where $E = \{(\rho, \theta, \phi) \mid 0 \leq \rho \leq 2, 0 \leq \theta \leq \pi/2, \pi/3 \leq \phi \leq \pi/2\}$ using the triple integral in spherical coordinates.

$$\begin{aligned} \iiint_E \sqrt{x^2+y^2+z^2} dV &= \int_{\pi/3}^{\pi/2} \int_0^{\pi/2} \int_0^2 \rho \cdot \rho^2 \sin \phi d\rho d\theta d\phi \\ &= \int_{\pi/3}^{\pi/2} \int_0^{\pi/2} \sin \phi \left[\frac{\rho^4}{4} \right]_{\rho=0}^{\rho=2} d\theta d\phi \\ &= \int_{\pi/3}^{\pi/2} \int_0^{\pi/2} 4 \sin \phi d\theta d\phi \\ &= \int_{\pi/3}^{\pi/2} 2\pi \sin \phi d\phi = 2\pi [\cos \phi]_{\phi=\pi/3}^{\phi=\pi/2} \\ &= \pi \end{aligned}$$